Enhancing Saudi Arabian Museums with Virtual Reality: Identifying the Best Technologies for Cultural Preservation and Engagement

Amani Albarazi
Computer Science, Effat College of Engineering
Effat University
Jeddah, Saudi Arabia
aalbarazi@effat.edu.sa

Raghad Alamoudi
Computer Science, Effat College of Engineering
Effat University
Jeddah, Saudi Arabia
rahalamoudi@effat.edu.sa

Maram Alhusami
Computer Science, Effat College of Engineering
Effat University
Jeddah, Saudi Arabia
maralhusami@effat.edu.sa

Tayeb Brahim
Effat Energy Engineering and Technology Research Center
Effat University
Jeddah, Saudi Arabia
tbrahimi@effatuniversity.edu.sa

***Abstract*—Virtual reality (VR) is a powerful tool for simulating environments and creating immersive experiences. . This study identifies the most suitable VR technologies for implementation in Saudi Arabian museums. Through an analysis of the state-of-the-art existing technologies and stakeholder consultations, the study highlights the potential of VR in enhancing cultural preservation and visitor engagement. A bibliometric analysis of research trends highlights the increasing prominence of advanced technologies such as human-computer interaction, the metaverse, extended reality, and 3D scanning, pointing to future developments in immersive and data-driven museum experiences. The research identified four key VR technologies: 3D scanning and rendering, full-immersion VR systems, AI-enhanced VR, and gesture recognition and motion tracking. These technologies offer immersive and interactive experiences that align with Saudi Vision 2030's emphasis on cultural and digital transformation. Additionally, the study supports Sustainable Development Goal (SDG) 9, Industry, Innovation, and Infrastructure, and SDG 11, Sustainable Cities and Communities, addressing both cultural heritage and technological advancement. The findings provide valuable insights and recommendations for integrating VR into Saudi museums to achieve innovation and engagement goals.**

Keywords—Virtual Reality, Saudi Arabian Museums, Visitor Interaction, Learning Experiences, Educational Value

I. INTRODUCTION

In the last decade, virtual reality (VR) has gained popularity in several fields such as training, industry, and medicine [1]. VR has also been widely accepted and well-received in recent years in cultural heritage, where it is used in several creative ways [2]. In particular, museums use VR to create immersive experiences that let visitors interact more deeply with art and history. With the ability of VR to simulate the historical environment and situation, it enables visitors to travel back in time and experience how events, places, and civilizations were in the past. By using storytelling, these historical reconstructions offer visitors rich, interactive narratives that improve their understanding of the past. The image in Fig.1, created using Dall-E (https://openai.com/index/dall-e-3/) illustrates a modern application of VR in museums, where visitors can engage with virtual reconstructions of artifacts and historical sites in an immersive setting. Such applications showcase how VR can transform static exhibits into dynamic, interactive experiences.

The first examples of VR in museums were simple virtual tours dating from the early 1990s [3]. As technology developed, the function of VR grew incredibly, opening the door to more sophisticated uses such as interactive displays and 3D reconstructions. Starting with virtual tours and reconstructions of the historic sites in the Louvre and British Museum, both large museums had integrated the latest VR into their processes by the 2010s. These new means of showing exhibitions to on-site and remote visitors allowed for novel ways of visiting them [4].

Fig. 1. Modern application of VR in museums (DALL-E-3).

Since Saudi Arabian museums are closely related to preserving the nation's rich cultural heritage, their embracing VR applications may also be seen as a strong opportunity for regional growth. Saudi museums have the potential to meet national goals in the areas of cultural preservation and education while also attracting younger and more technology-savvy tourists by using VR to create immersive surroundings and historical narratives. However, while museums play a crucial role in preserving cultural heritage, they struggle to engage visitors, especially younger, tech-savvy audiences [9]. Traditional exhibitions often lack the interactivity that modern consumers seek Arabia's Vision 2030 emphasizes cultural development and creating opportunities to enhance museum experiences [7]. VR can offer exciting new ways for visitors to engage with exhibits[10]. Some countries have already embraced VR, but Saudi Arabia has yet to fully leverage its potential, making it a promising tool to improve educational experiences and visitor engagement in local museums [11]. The research question of this paper is: what are the best VR technologies for Saudi Arabian museums? the research objectives are first to investigate and analyze different VR technologies applied in museums worldwide, identify current gaps or limitations in applying VR technologies in museums, propose the most appropriate VR technology to be implemented in Saudi Arabian museums, and finally offer recommendations for future research directions in the realm of VR technologies for Sadi Arabian museums. The present study supports Saudi Arabian Vision 2030 Development of Cultural and Entertainment Sectors Goal 1.1.2 as well as the UN Sustainable Development Goals (SDGs), namely SDG 9 (Industry, Innovation, and Infrastructure), Target 9.5 and SDG 11 (Sustainable Cities and Communities), Target 11.4, protecting the world's cultural and natural heritage https://sdgs.un.org/goals.

The structure of this paper is as follows: Section 2 provides a comprehensive literature review on VR and museums. Section 3 details the research methodology used in this study. Section 4 presents, analyzes and discusses the findings. Finally, Section 5 presents a conclusion and recommendations for future work.

II. Literature Review

VR is increasingly being integrated into cultural institutions, particularly museums. Fig 2. illustrates the number of publications on VR in museums from 2000 to 2024 in the Lens database. This upward trajectory in scholarly output highlights the growing recognition of VR's potential to enhance museum experiences.

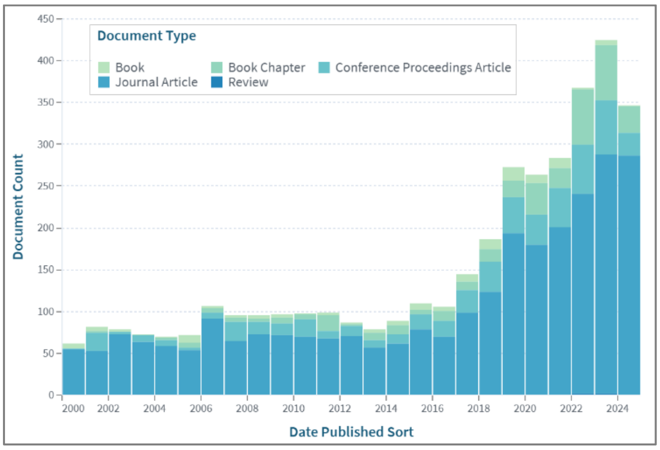

Fig. 2. Number of publications on Virtual Reality in Museums from 2000 to 2024 in Lens database.

The search strategy and methodology for identifying relevant literature involves searching through databases such as Google Scholar, Lens.org, IEEE, Web of Science, and others. Search terms are combined with specific inclusion and exclusion criteria to refine the search results, particularly focusing on articles published within the last 10 years. After filtering, we reviewed the remaining papers and selected those that best aligned with our study's objectives. This process is followed by a synthesis of the existing literature and a critical evaluation of the solutions, approaches, or theories discovered, assessing their effectiveness. Finally, the review concludes with the identification of gaps in existing knowledge, pinpointing any inconsistencies or areas for further research within the existing body of knowledge on the topic.

Key technologies identified in the literature show a variety of VR tools adopted by museums globally to enhance visitor engagement and cultural preservation. The technologies range from AI-enhanced VR, which personalizes experiences and generates dynamic narratives, to gesture recognition and motion tracking, which enables interactive user engagement. To create detailed virtual replicas of artifacts, bridging the gap between physical and digital experiences #D tools like 3D photogrammetry and 3D scanning and rendering were adopted. Others used AI and crowdsourcing to enrich cultural metadata, like platforms CrowdHeritage or the 2gether VR Museum. These technologies have been implemented in prominent museums such as the Louvre, British Museum, and National Museum of Cinema, showcasing their transformative potential. Table 1 provides a detailed overview of these technologies, their applications, and examples museums of and institutions that adopted them.

TABLE I. Technologies Used in Museums

Technologies	Description	Museums	Ref
Artificial Intelligence Enhanced VR	AI algorithms can develop in VR through personal experience, dynamic narrative generation, and object recognition automation.	The Natural History Museum in the UK	[13], [8], [14], [15]
The CrowdHeritge Platform	An ecosystem using AI and crowd-sourcing for the enrichment of cultural heritage metadata, enabling the annotation of material by users, thus increasing public participation.	Rijksmuseum in the Netherlands	[10], [16], [17]
Computational Methods	Various mathematical, algorithmic, and computational techniques used to create, process, and enhance virtual environments and interactions within a VR system.	Louvre Museum in France	[18], [19]
Gesture Recognition and Motion Tracking	Sensors and cameras track the movement of users, translating movements into interactions within the VR environment.	The National Museum of Natural History in France.	[20], [21], [22], [23]
3D Photogrammetry	Increases Museum Engagement through developing highly detailed 3D models for immersive exploration to close the experience gap between digital and physical.	British Museum in London and the Louvre Museum in France	[24], [3]
3D Scanning and Rendering	3D scanning captures real-world objects or environments and digitizes them into 3D models. Such models can be integrated into VR platforms for highly detailed, accurate virtual experiences.	The National Museum of Anthropology in Mexico	[25], [13], [26], [27]
2gether VR Museum	The platform allows users to create virtual environments where visitors can walk through, view artifacts in context, and engage with interactive visuals, animations, and activities.	The platform is still experimental as of yet.	[6], [28]

Full-Immersion VR Systems	These systems immerse users in a completely virtual environment utilizing HMDs, hand controllers, and sometimes haptic feedback devices.	The National Museum of Cinema in Italy	[8], [29], [30], [31]

The integration of AI and human crowdsourcing significantly enhances metadata quality and searchability in cultural heritage, effectively engaging users in the annotation process [18][16]. This approach allows for precise enrichment of cultural content [17]. VR offers substantial benefits for museums, enhancing visitor engagement through immersive experiences and facilitating virtual access to historically significant sites [1][10]. While VR increases educational value and accessibility, it also poses financial challenges related to implementation and personnel training. Balancing virtual and physical exhibits is crucial to ensure that virtual experiences do not overshadow actual artifacts.

Computational tools provide insights by analyzing visual patterns in photographs [32][18]. The 2gether VR platform addresses challenges in virtual experiences with features like teleportation navigation and interactive tools, enhancing understanding of modern art [28]. Combining VR with photogrammetry can create immersive exhibitions, though factors like photo quality are essential, especially for Saudi museums [24][6].

3D scanning and rendering technologies, as seen in the National Museum of Anthropology in Mexico, preserve cultural artifacts and create detailed virtual models, enhancing global accessibility and engagement [27][26][13][25]. Full-immersion VR Systems, utilizing Head-Mounted Displays and haptic feedback, provide engaging experiences that allow users to interact dynamically with exhibits, as demonstrated in museums like the National Museum of Cinema in Italy [8][31][29][30].

A critical review of the literature has revealed some gaps regarding the use of VR within a museum context. Each of these identifies some areas of adoption-related logistics and technology issues to be resolved for full usage of VR technology in museums to occur. These gaps are summarized in Table II. The gaps point to areas that require more study and advancement in the application of VR technology in museums.

TABLE II. GAPS IDENTIFIED IN VR IMPLEMENTATION IN MUSEUMS

Gaps	Description	Ref
AI Limitations	Automated tools have difficulty with domain-specific content, as most of the AI models are trained outside the cultural heritage domain, hence giving in correct results	[16], [17], [33]
Metadata Consistency	To date, metadata standardization and consistency across different cultural platforms are not yet optimized processes. Skepticism and Adoption Due to their novelty and the perceived threat to traditional expertise, art historians are sometimes skeptical about computational methods.	[24], [34]
Long-term Impact of VR on Visitor Behavior	There are hardly any studies on sustained visitor engagement or learning outcomes after being exposed to VR.	[1], [35]
Cost-effectiveness of VR Installations	Lack of enough research on long-term financial viability regarding the setup of VR in museums.	[28], [36], [10]
Integration of VR and Traditional Artifacts	Limited research on how best to balance the use of VR with physical artifacts in museum curation.	[1], [32]
User Familiarity with VR	The target audience's lack of familiarity with VR could be a limiting factor regarding engagement.	[3], [10]
Image Quality Dependency	Success in digitization is highly dependent on source image quality.	[10], [17], [24]
Complex Artifact Handling	Inability to capture accurately complex shapes or delicate specimens.	[24], [37]

III. METHODOLOGY

The research method used in this study involves a combination of state-of-the-art analysis and stakeholder consultation to analyze and identify the most suitable VR technologies for Saudi Arabian museums along with a bibliometric analysis using VOSviewer. The state-of-the-art analysis systematically searched different databases, like Google Scholar, lens.org, Scopus, and the Web of Science Database focusing on publications in the last 10 years. The inclusion and exclusion criteria were applied to refine the results, their relevance, and their credibility. The query used in this study is: (("VIRTUAL REALITY" OR "VR") AND "MUSEUM*").

We decided to implement the VR for Abdul Raouf Khali House, which is a part of the museum, using the recommended programming language Unity for VR development, adding interactivity to enhance user engagement, and creating scenarios that simulate real museum experiences. Stakeholder consultations were conducted through discussions and interviews to gather valuable feedback and perspectives, enriching the understanding of user needs and expectations. This mixed-method approach enabled the study to balance innovation, practicality, and cultural relevance in selecting VR technologies.

IV. RESULTS AND DISCUSSION

We aimed to analyze and select the potential VR technologies to implement in Saudi Arabian museums. We reviewed and analyzed different types of papers with different objectives to reach our aim. This section of the results and discussion presents the main results found based including a bibliometric analysis using the Web of Science database.

Using VOSviewer, Figure 3 presents a detailed visualization of the published work on VR technologies in museums for enhancing visitor engagement and cultural preservation, using the VOSviewer tool with the query: (("VIRTUAL REALITY" OR "VR") AND "MUSEUM*"). The visualization from the Web of Science database highlights key themes and their interconnections within VR and museums and cultural heritage such as such as virtual reality, cultural heritage, and museums, which are the most dominant keywords in the network indicating their importance as the primary focus areas. On the emerging technologies, we found 3D scanning, 3D visualization, 3D printing, and mixed reality which indicates their contribution to virtual heritage and interactive museums. In other related areas, we can find human-computer interaction, digital reconstruction, and machine learning which highlights the multidisciplinary nature of VR applications in museums. The overall interconnected network demonstrates the increasing interest on VR applications in museums to enhance cultural preservation, accessibility, and visitor engagement.

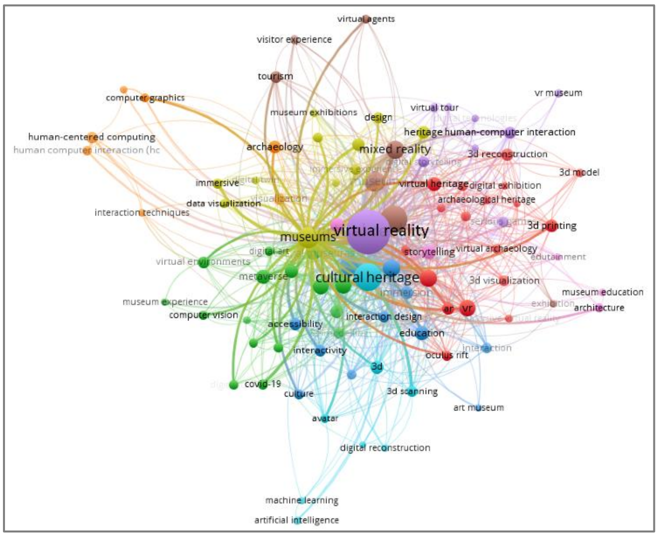

Fig. 3. Keyword co-occurrence visualization based on WoS database.

An interesting result using overlay visualization is shown in Figure 4. This figure highlights the evolution of research trends in VR and cultural heritage from 2019 to 2022. Results (red spot) indicate an increasing focus on advanced technologies such as human-computer interaction, metaverse, VR, extended reality, immersive technology, digital reconstruction, machine learning, and 3D scanning, these areas signify potential future directions for research pointing to future developments in creating immersive and data-driven museum experiences.

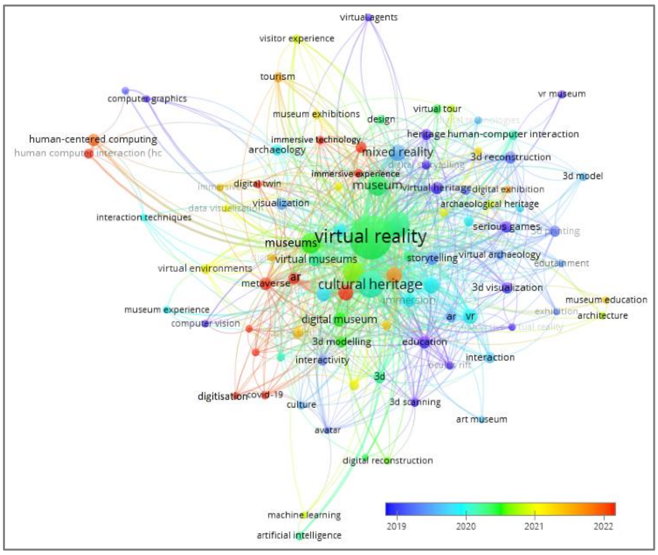

Fig. 4. VR and cultural heritage trend overlay visualization.

Based on the existing VR technologies globally, we have identified different technologies that could be applied to Saudi Arabian museums. We found 4 of the VR technologies that support our goal to identify the best VR technologies that provide visitors with an interactive environment, a learning experience, and educational value, namely 3D Scanning and Rendering, Full-Immersion VR Systems, Artificial Intelligence-Enhanced VR, and Gesture Recognition and Motion Tracking. These technologies are summarized in Table III.

TABLE III. BEST VR TECHNOLOGIES FOR SAUDI ARABIAN MUSEUMS

Technology	Description	Ref
3D Scanning and Rendering	3D scanning captures real-world objects or environments and digitizes them into 3D models. Such models can also be integrated into VR platforms for highly detailed, accurate virtual experiences	[25], [13], [26], [27]
Full-Immersion VR Systems	These systems immerse users in a completely virtual environment utilizing HMDs, hand controllers, and sometimes haptic feedback devices.	[8], [29], [30], [31]
Artificial Intelligence-Enhanced VR	AI algorithms can develop in VR through personal experience, dynamic narrative generation, and object recognition automation.	[13], [8], [14], [15]
Gesture Recognition and Motion Tracking	Sensors and cameras track the movement of users, which then translate movements into interactions within the VR environment.	[20], [21], [22], [23]

The analysis highlights key aspects of applying VR technologies in Saudi Arabian museums, focusing on both their potential and challenges for enhancing visitor engagement, learning, and cultural preservation. The 3D Scanning and Rendering enable the creation of high-fidelity digital replicas of artifacts, providing immersive and accurate representations that traditional displays cannot offer [25], [13]. Full-immersion VR Systems transform passive viewing into active exploration, enriching the learning experience by immersing users in virtual environments [8], [2]. Artificial Intelligence-Enhanced VR personalizes content to users' preferences and learning styles, improving engagement and interactivity and enhancing educational outcomes [13],[8]. Finally, Gesture Recognition and Motion Tracking allow users to engage with exhibits through natural movement, increasing participation and promoting deeper interaction with the exhibits [20], [21]. These technologies collectively have the potential to revolutionize museum experiences by creating more interactive, immersive, and educational environments.

While this study identified four key VR technologies, as particularly relevant to Saudi Arabian museums, several other VR technologies were reviewed but ultimately not selected for implementation due to specific limitations or challenges. The CrowdHeritage Platform uses AI and crowdsourcing to enrich cultural heritage metadata, but given its reliance on user input, this raised concerns about data quality and authenticity, issues considered crucial in museum contexts found in [10], [16]. Similarly, while Computational Methods can produce advanced simulations, their technical requirements and high computational demands made them less practical for Saudi museums as well [18],[19]. 3D photogrammetry allows the capture of detailed 3D models to enable immersive exploration, but because the technique requires a great deal in equipment and expertise, broader applicability is out of reach currently [24],[3]. The 2gether VR Museum platform has emerged, demonstrating great promise for generating virtual environments, but its complete development is still at the prototype stages, which puts off any large-scale application or uses thereof [6],[28]. These technologies, though worthy in particular contexts, have not been found best suited to the immediate needs of this study, which aimed to prioritize solutions balancing innovation, practicality, and cultural relevance.

Future research should focus on developing culturally appropriate VR content, conducting long-term studies on the impacts and sustainability of VR in museums, and fostering collaboration between technology experts, cultural stakeholders, and academic institutions.

V. Conclusion

This study analyzed and selected the best VR technologies to be implemented in Saudi Arabian museums focusing on enhancing visitor engagement, learning experiences, and cultural preservation. The bibliometric analysis revealed a growing research interest in VR applications in museums, with recent trends (2019–2022) highlighting advanced technologies such as human-computer interaction, the metaverse, extended reality, digital reconstruction, machine learning, and 3D scanning. Among the key findings are the identification of four promising technologies: motion tracking and gesture recognition, full-immersion VR systems, AI-enhanced VR, and 3D scanning and rendering. These have indeed been shown to provide an interactive experience. While supporting the SDGs and Saudi Vision 2030. However, challenges such as high costs, the need for culturally sensitive information, and a lack of long-term sustainability data were highlighted.

Future research should also be done to create appropriate VR content for the culture of Saudi Arabian viewers to ensure effective engagement. Long-term research regarding the effects and sustainability of VR technologies in museums should be carried out to conceptualize their potential in education and the preservation of culture.

Acknowledgment

The authors gratefully acknowledge the support from the College of Engineering at Effat University, Jeddah, Saudi Arabia. The authors also acknowledge the valuable discussion, suggestions, and support received from Passent Elkafrawy a doctor of Effat College of Engineering at Effat University, Jeddah, Saudi Arabia.

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